

Supplementary Material

Adapter Scaling Trade-off and Timestep-Conditioned Scheduling in Multi-view Diffusion Texture Generation

FAC controlled re-examination: full dose-response study

1 S1. Protocol

All variants are trained and evaluated under the strictly paired protocol of Sec. 4.6 of the main manuscript: the base model, VAE, and GeoTex-Adapter remain frozen; only the LTAG/GSG/FSC controllers (roughly 6×10^3 parameters) are trained with the same enhanced denoising objective as the base adapter (a latent noise-prediction MSE with $1.7\times$ foreground and $1.3\times$ edge spatial reweighting plus a latent-SSIM term; AdamW, peak learning rate 10^{-4} with 100 warm-up steps), on the adapter’s training pool, which is strictly disjoint from all evaluation objects; evaluation uses 300 objects, the Euler discrete scheduler with 50 steps, seed 42, and identical evaluation code for every variant. Deficits are paired per object against the training-free TCAS baseline ($C3 = (1.25, 2.50, 1.25)$).

2 S2. Dose-response over seven trained configurations

Table 1: FAC controlled re-examination: seven trained configurations, all evaluated under the strictly paired protocol. FG-PSNR and FG-SSIM are foreground-masked absolute values; Δ FG-PSNR is the paired mean difference relative to the training-free TCAS baseline; the win rate is the fraction of objects on which the variant improves foreground PSNR. The untrained warm-start control (12-object paired subset, gate fixed at the exact C3 envelope) is reported in the text: Δ FG-PSNR = -0.18 dB, 95% CI $[-0.42, +0.05]$, statistically indistinguishable from the baseline.

Configuration	Temporal gate	FG-PSNR	FG-SSIM	Δ FG-PSNR	Win rate
TCAS baseline (C3)	fixed C3	9.88	0.3681	–	–
LTAG only	joint, unconstrained	7.79	0.3487	–2.08	4/300
LTAG+GSG	joint, unconstrained	7.03	0.3163	–2.84	6/300
Full FAC	joint, unconstrained	6.94	0.3204	–2.94	5/300
Two-stage + GSG/FSC	frozen at C3	7.32	0.3074	–2.56	4/300
+ envelope penalty $\lambda=1$	joint, penalized to C3	7.14	0.3021	–2.74	2/300
+ fidelity weighting	frozen at C3	7.51	0.3110	–2.37	5/300
+ gentle schedule	frozen at C3	8.77	0.4175	–1.10	48/300

3 S3. Reading of the dose-response pattern

Three observations follow from Table 1. First, the warm-start control shows that when the temporal gate is held at the exact C3 envelope and not trained, the extension is statistically indistinguishable from the training-free schedule (paired $\Delta = -0.18$ dB, 95% CI $[-0.42, +0.05]$, crossing zero); the evaluation path is therefore validated. Second, the deficit grows monotonically with the amount of controller training under the denoising objective: zero steps preserve the baseline, gentle training (500 steps at a fourfold-reduced learning rate) costs 1.10 dB, and full training costs up to 2.94 dB. Third, envelope preservation is necessary but not sufficient: even with the gate frozen at C3, training the spatial and frequency controllers degrades fidelity, and the gentle variant merely trades signal fidelity for a slightly higher structural statistic (FG-SSIM $+0.049$ at FG-PSNR -1.10 dB) rather than achieving a genuine improvement.

The mechanism is consistent with the controllers optimizing the denoising objective at the expense of foreground fidelity: gradient training on the same loss as the base adapter has no term that protects the fixed schedule’s balance, so the learned modulation migrates away from the operating point that makes TCAS effective. We therefore identify, as future work, training signals that do not trade away foreground fidelity (fidelity-weighted or structure-aware objectives), envelope-preserving parameterizations and constraints, and post-hoc rather than gradient-learned gates. With the tested lightweight controllers, the training-free TCAS remains the main and sole method of the paper, and all its conclusions are unaffected by this negative result.